DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& also for CS



Calculus and Optimization

Functions and Graphs

Lecture No. 02



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Recap of previous lecture



Topic

Basics of Graphs (I)

Topics to be Covered



Topic

Basics of Graphs (II)

INFORMATION

Types of functions

- ALGEBRAIC function
  - (1) Polynomial func^n
  - (2) Rational func^n
  - (3) Irrational func^n
  - (4) Piecewise func^n
    - Mod func^n
    - Signum func^n
    - G.I.f.
    - L.I.f
    - F.P.f
- TRANSCEDENTAL function
  - (1) Exponential func^n
  - (2) log function.
  - (3) Trigonometric func^n
  - (4) Inverse Trig. functions.

G.I.f = Greatest Integer Func^n (Floor func^n)
L.I.f = Least Integer func^n (Ceiling func^n)
F.P.F = Fractional Part Func^n

PIECEWISE func^n → If func^n is defined by Multiple sub function

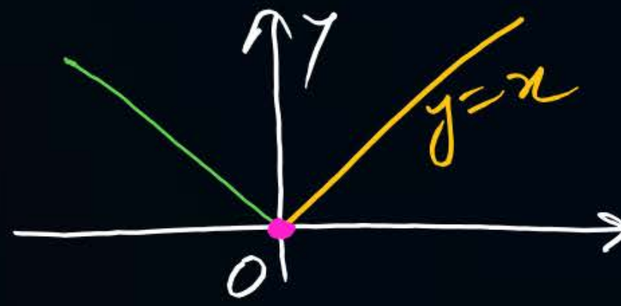
s.t, Domain of each subfunction is different

then function is called Piecewise func^n

for eg. Mod func^n, Signum func^n, G.I.F, L.I.F, Fractional Part func^n etc

MODULUS func^n →

$$y = |x| = \begin{cases} -x, & x < 0 \\ +x, & x > 0 \end{cases}$$



$$= \max\{x, -x\}$$

eg $|5| = +5$, $|-5| = -(-5) = +5$

eg $|5| = \max\{5, -5\} = 5$

$|-5| = \max\{-5, -(-5)\}$

$= \max\{-5, +5\} = +5$

special Def^n of $|x|$ → $y = |x| = \sqrt{x^2}$

eg $|5| = \sqrt{(5)^2} = \sqrt{25} = +5$

& $|-5| = \sqrt{(-5)^2} = \sqrt{25} = +5$

Q if $x^2 - 25 = 0 \Rightarrow x = ?$

(M-I) $(x-5)(x+5) = 0 \Rightarrow x = -5$ or 5

(M-II) $x^2 - 25 = 0 \Rightarrow x = \sqrt{25} = \pm 5$

WRONG Explanation.

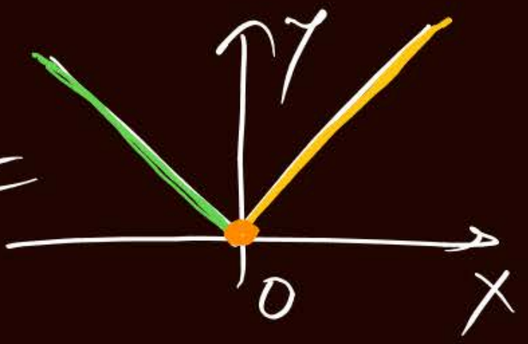
(M-III) $x^2 - 25 = 0 \Rightarrow x^2 = 25$

$x = \pm(\sqrt{25})$ ↓ Correct Version.

$= \pm(+5)$

$x = \pm 5$

Modulus func^n: $y = |x| = \sqrt{x^2} = \begin{cases} -x, & x<0 \\ x, & x>0 \end{cases} = \max(x, -x) =$



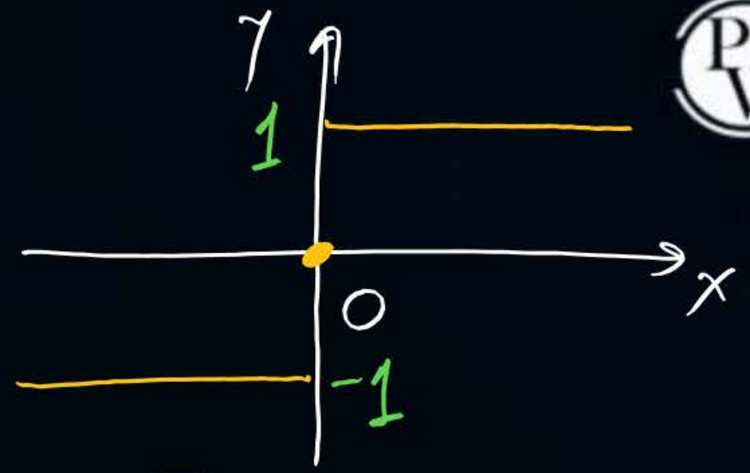
① It is an (Even) func^n

② Domain $(x) = (-\infty, \infty) = R$

③ Range $(y) = [0, \infty) = R^+$

⊛ Signum func^{n} →

$$y = \operatorname{sgn}(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \\ 0, & x = 0 \end{cases}$$



Domain$(x) = (-\infty, \infty)$, Range$(y) = \{-1, 0, 1\}$, ODD function

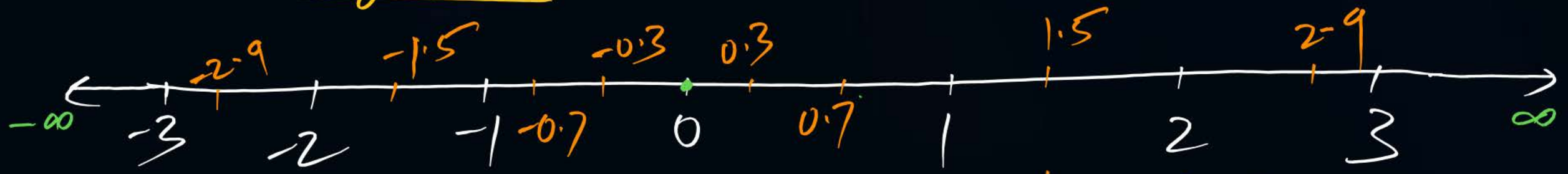
eg $\operatorname{sgn}(0) = 0$, $\operatorname{sgn}(1.3) = 1$, $\operatorname{sgn}(1.9) = 1$, $\operatorname{sgn}(2.7) = 1$ ----

& $\operatorname{sgn}(-3.4) = -1$, $\operatorname{sgn}(-0.2) = -1$, $\operatorname{sgn}(-4.5) = -1$ ----

$\operatorname{sgn}(1) = 1$, $\operatorname{sgn}(2) = 1$, $\operatorname{sgn}(3) = 1$ ------

$\operatorname{sgn}(-1) = -1$, $\operatorname{sgn}(-2) = -1$, $\operatorname{sgn}(-3) = -1$

(*) Greatest Integer func^n (Floor function) →



$\lfloor 0.3 \rfloor = 0$, $\lfloor 0.7 \rfloor = 0$, $\lfloor 1.5 \rfloor = 1$, $\lfloor 2.9 \rfloor = 2$ .....

$\lfloor -0.3 \rfloor = -1$, $\lfloor -0.7 \rfloor = -1$, $\lfloor -1.5 \rfloor = -2$, $\lfloor -2.9 \rfloor = -3$ .....

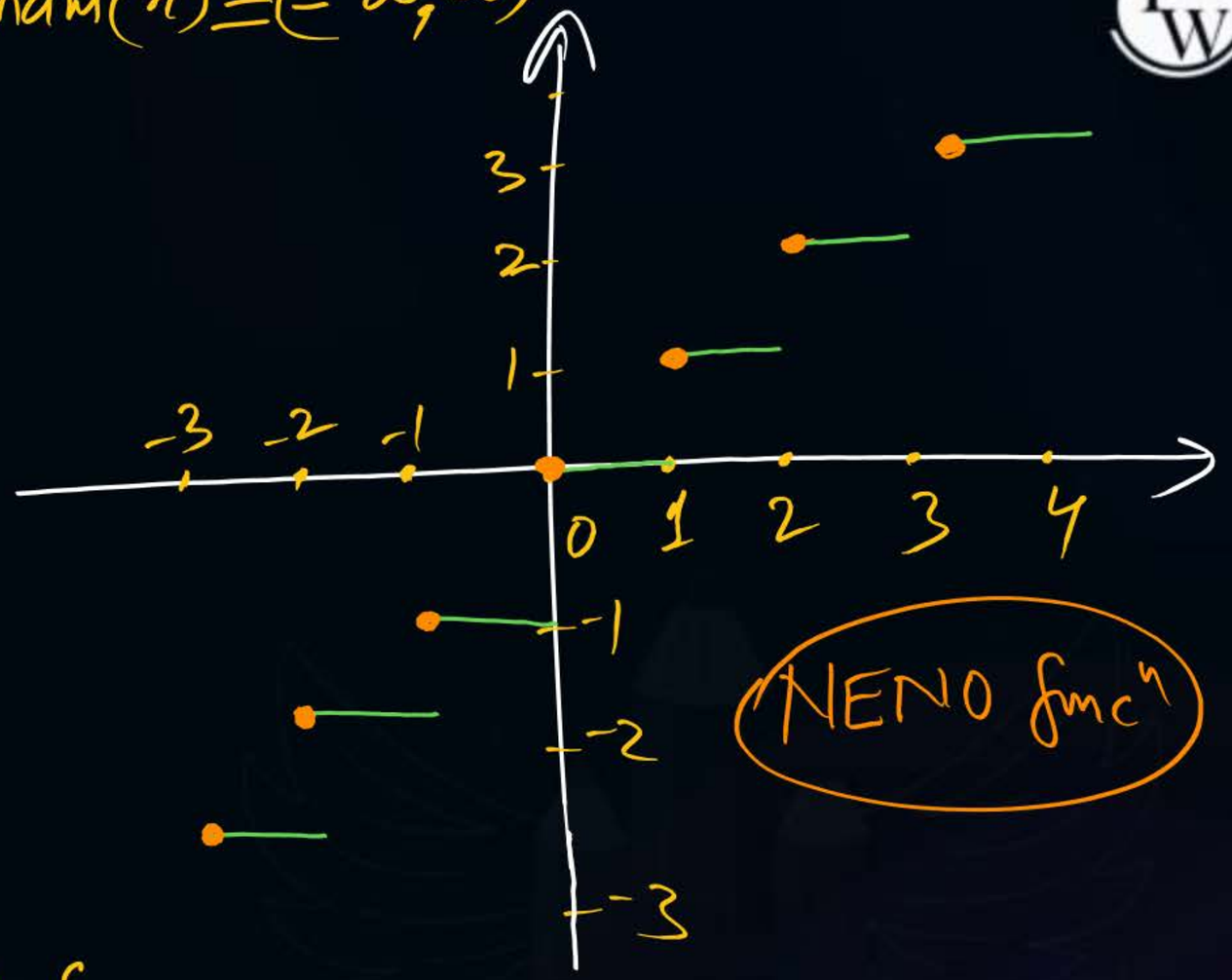
$\lfloor 0 \rfloor = 0$, $\lfloor 1 \rfloor = 1$, $\lfloor 2 \rfloor = 2$, $\lfloor 3 \rfloor = 3$ .....

$\lfloor -1 \rfloor = -1$, $\lfloor -2 \rfloor = -2$, $\lfloor -3 \rfloor = -3$ .....

Graph of G.I.F → (Floor fun^{n})

Domain(x) = $(-\infty, \infty)$

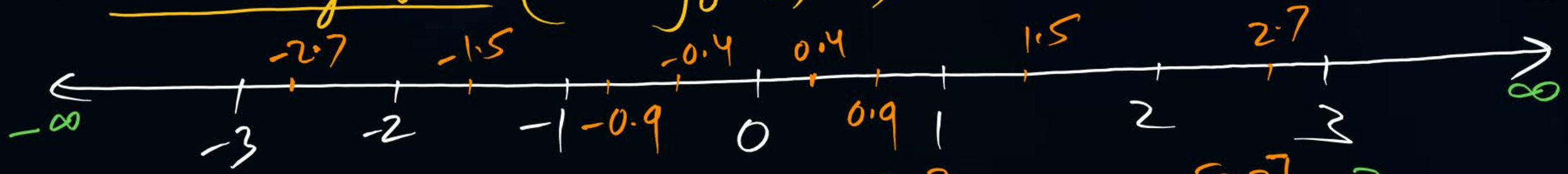
$$y = [x] = \begin{cases} -3, & -3 \le x < -2 \\ -2, & -2 \le x < -1 \\ -1, & -1 \le x < 0 \\ 0, & 0 \le x < 1 \\ 1, & 1 \le x < 2 \\ 2, & 2 \le x < 3 \\ 3, & 3 \le x < 4 \end{cases}$$



NENO fun^{n}

Range(y) = {....-3, -2, -1, 0, 1, 2, 3....} = Set of Integers

⊛ Least Integer func^n (ceiling func^n) →



$\lceil 0.4 \rceil = 1$, $\lceil 0.9 \rceil = 1$, $\lceil 1.5 \rceil = 2$, $\lceil 2.7 \rceil = 3$ .....

$\lceil -0.4 \rceil = 0$, $\lceil -0.9 \rceil = 0$, $\lceil -1.5 \rceil = -1$, $\lceil -2.7 \rceil = -2$ ....

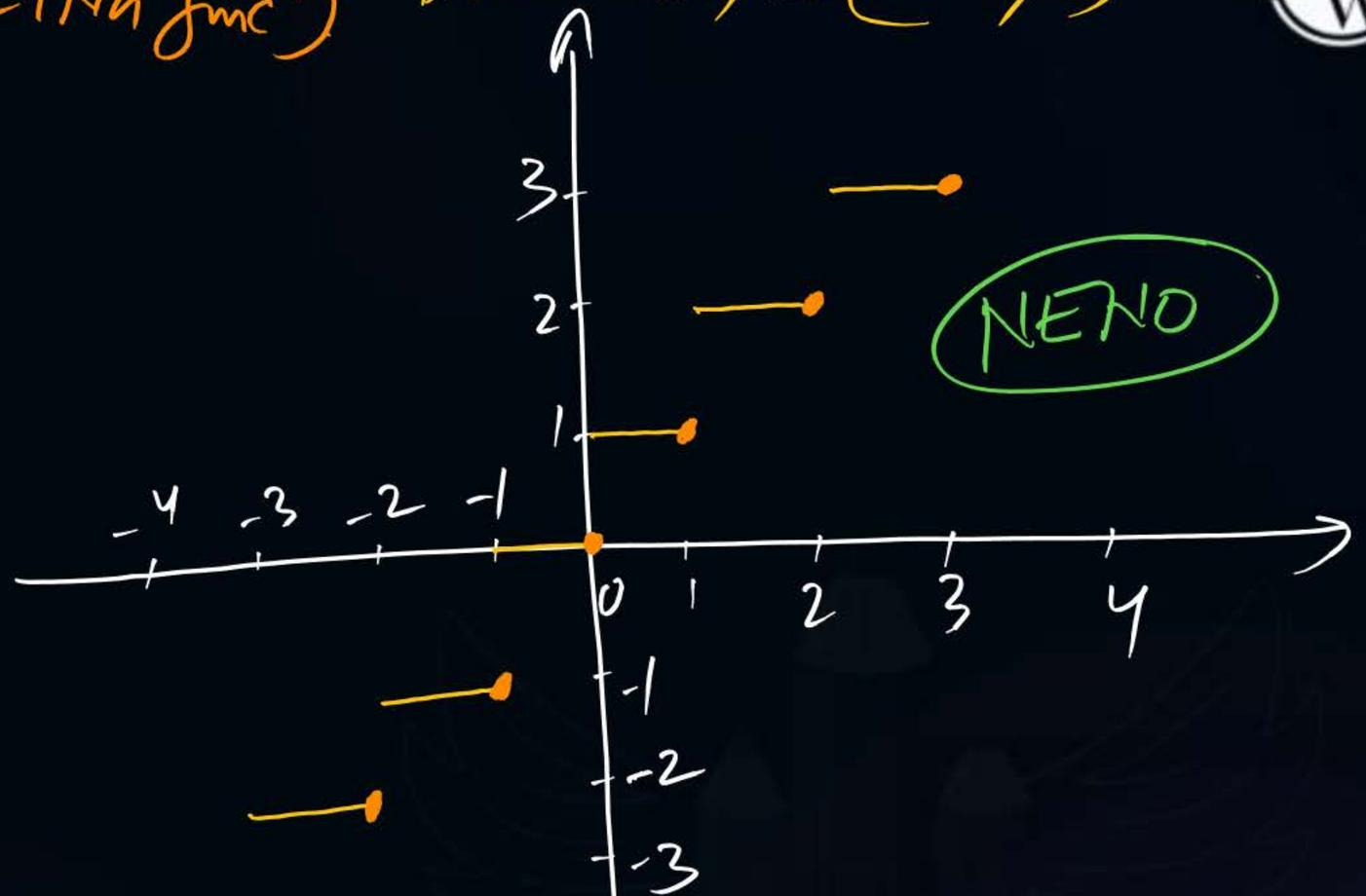
$\lceil 0 \rceil = 0$, $\lceil 1 \rceil = 1$, $\lceil 2 \rceil = 2$, $\lceil 3 \rceil = 3$ .....

$\lceil -1 \rceil = -1$, $\lceil -2 \rceil = -2$, $\lceil -3 \rceil = -3$ .....

Least Integer function → (CEILING func^n)

Domain $(x) = (-\infty, \infty)$

$$y = \lceil x \rceil = \begin{cases} -1, & -2 < x \le -1 \\ 0, & -1 < x \le 0 \\ 1, & 0 < x \le 1 \\ 2, & 1 < x \le 2 \\ 3, & 2 < x \le 3 \\ 4, & 3 < x \le 4 \end{cases}$$



NEND

Range $(y) = \{\ldots -2, -1, 0, 1, 2, 3, \ldots\}$ = Set of Integers.

Fractional Part func^n : $y = \{x\} = x - \lfloor x \rfloor$

$\{0.2\} = 0.2$, $\{0.7\} = 0.7$, $\{1.3\} = 0.3$, $\{2.6\} = 0.6$ ------

$\{-0.2\} = (-0.2) - \lfloor -0.2 \rfloor$
$= -0.2 - (-1)$
$= 0.8$

$\{-0.7\} = (-0.7) - \lfloor -0.7 \rfloor$
$= -0.7 - (-1)$
$= 0.3$

$\{-2.6\} = (-2.6) - \lfloor -2.6 \rfloor$
$= -2.6 - (-3)$
$= -2.6 + 3$
$= 0.4$

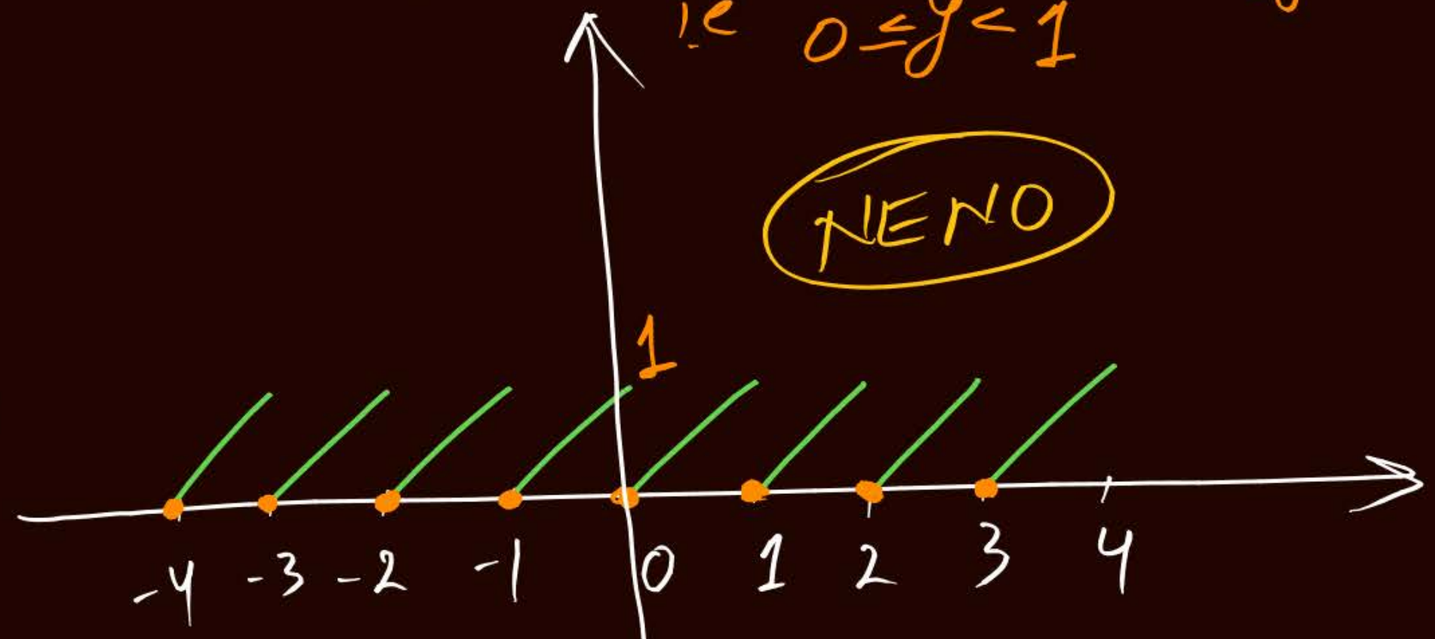
$\{-1.3\} = (-1.3) - \lfloor -1.3 \rfloor$
$= -1.3 - (-2)$
$= 0.7$

$\{0\} = 0$
$\{1\} = 0$, $\{-1\} = (-1) - \lfloor -1 \rfloor = -1 - (-1) = 0$
$\{2\} = 0$, $\{-2\} = 0$, $\{-3\} = 0$ .....

Graph of f.P.F: $y = \{x\} = x - [x]$ = It's Graph lies in b/n 0 & 1 in terms of y

i.e $0 \le y < 1$

$$y = \{x\} = \begin{cases} x+3, & -3 \le x < -2 \\ x+2, & -2 \le x < -1 \\ x+1, & -1 \le x < 0 \\ x, & 0 \le x < 1 \\ x-1, & 1 \le x < 2 \\ x-2, & 2 \le x < 3 \\ x-3, & 3 \le x < 4 \\ \vdots & \vdots \end{cases}$$



Domain $= (-\infty, \infty)$, Range $(y) = [0, 1)$

Polynomial: It's Domain is $(-\infty, \infty)$ & Degree $= 0, 1, 2, 3, 4, 5 \ldots$ ——.
& it's Definition is same at all points in the Domain of $y = f(x)$

eg $y = K$ (Constant Poly.) $\simeq$ degree $= 0$

$y = ax + b$ (Linear poly) $\simeq$ Deg $= 1$

$y = ax^2 + bx + c$ (Quad. poly) $\simeq$ Deg $= 2$

$y = ax^3 + bx^2 + cx + d$ (Cubic poly) $\simeq$ Dy $= 3$

⋮

Sp Note- $y = |x| = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases}$, $D_f = (-\infty, \infty)$, it a poly of degree $= ?$

$=$ Not a polynomial

BASIC GRAPHS :→

① Equⁿ of x axis : $y=0$

② " of line ∥ᵍ to x axis $\boxed{y=k}$ (Constant Poly = Horizontal line)
Degree = 0 = Even degree.

③ Equⁿ of y axis : $x=0$

④ Equⁿ of line ∥ᵍ to y axis $x=k$ (Vertical line)

⑤ Equⁿ of line passing through origin is $\boxed{y=mx}$ where $\boxed{m=\tan\theta}$ = Slope of line



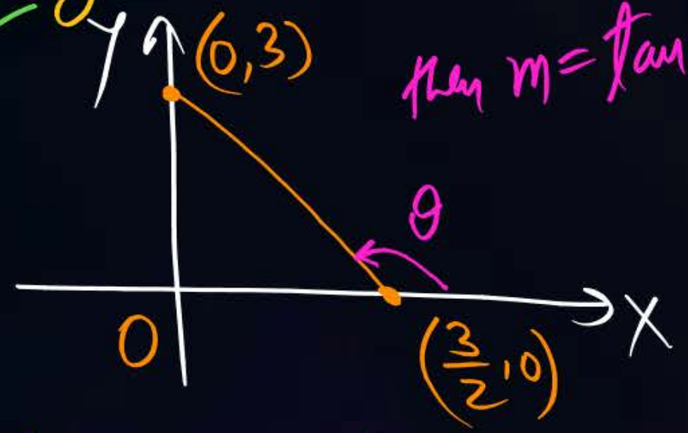
⑥ Slope Intercept Form of line:

$y = mx + c$ where

m = slope of line

c = Intercept cuts of y axis

eg $y = -2x + 3$



then $m = \tan\theta = -2$

At X axis, Put $y = 0$ then

$\rightarrow 0 = -2x + 3 \Rightarrow x = \frac{3}{2}$

⑦ Intercept Form of Line

$\frac{x}{a} + \frac{y}{b} = 1$ where a = X Intercept, b = Y Intercept.

eg $y = -2x + 3$

$2x + y = 3 \Rightarrow \frac{2x}{3} + \frac{y}{3} = 1 \Rightarrow \boxed{\frac{x}{3/2} + \frac{y}{3} = 1}$

eg: $2y = 4x - 8$

$-4x + 2y = -8 \Rightarrow \frac{-4x}{-8} + \frac{2y}{-8} = 1 \Rightarrow \boxed{\frac{x}{2} + \frac{y}{-4} = 1}$



Note: $2y = 4x - 8$

$\boxed{y = 2x - 4}$

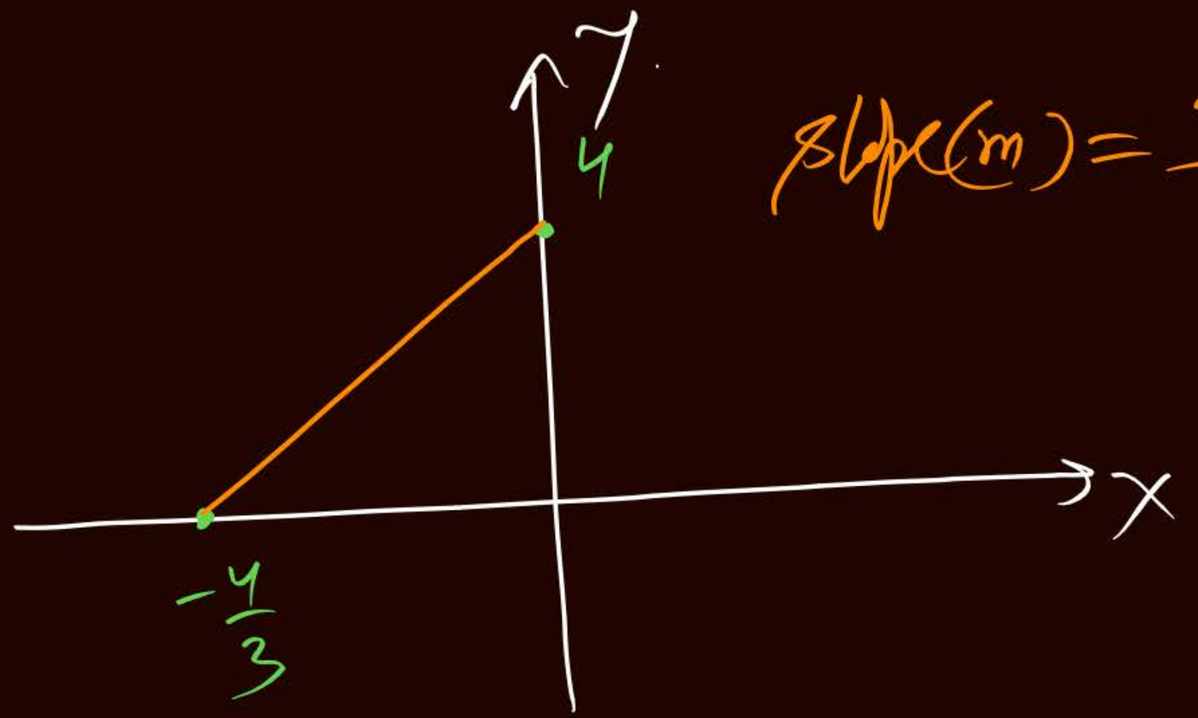
Slope $(m) = 2$

Eg: $y = 3x + 4$

$-3x + y = +4$

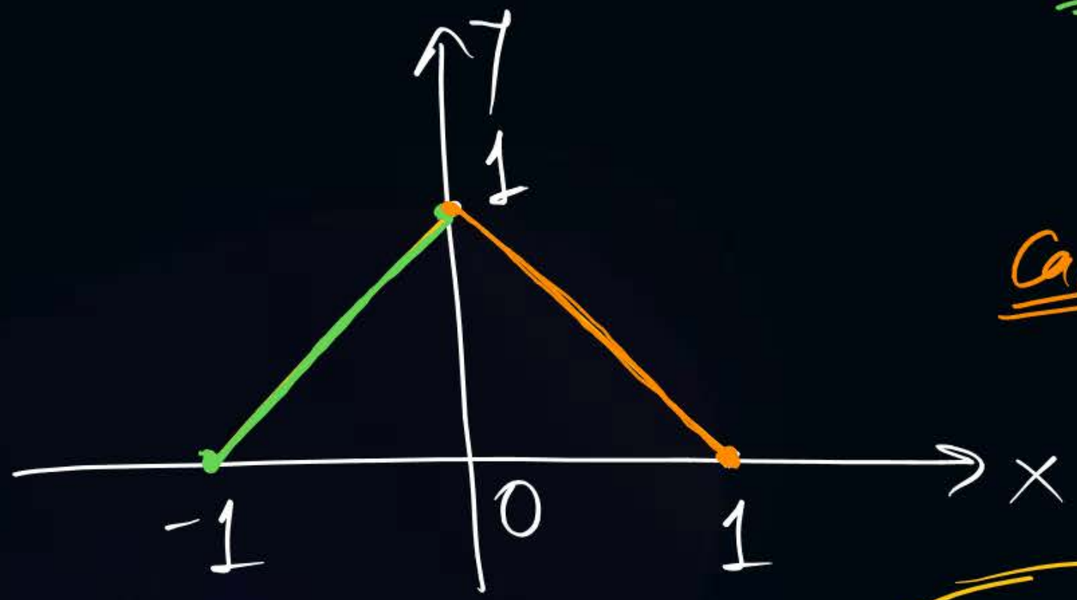
$\frac{-3x}{+4} + \frac{y}{+4} = 1$

$\frac{x}{-4/3} + \frac{y}{+4} = 1$



Slope (m) = 3

PQ: Draw the graph of $y = \begin{cases} 1+x, & -1 \le x \le 0 \\ 1-x, & 0 \le x \le 1 \end{cases}$



Case I: $-1 \le x \le 0$ then $y = 1+x$

i.e. $-x+y=1 \Rightarrow \frac{x}{(-1)} + \frac{y}{(1)} = 1$

Case II: if $0 \le x \le 1$ then $y = 1-x$

i.e. $x+y=1 \Rightarrow \frac{x}{(1)} + \frac{y}{(1)} = 1$

Domain $(x) = [-1, 1]$

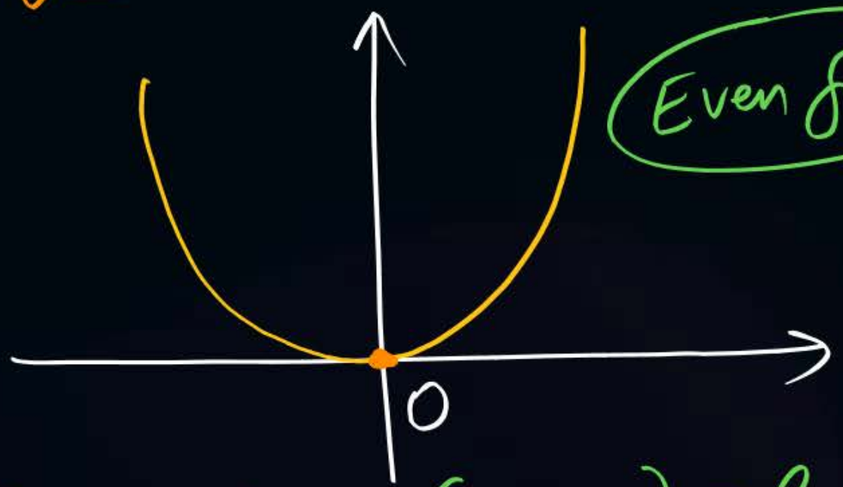
Range $(y) = [0, 1]$

Even fun^{n}

⊛ Poly of Even Degree

INFO

$y = x^2, x^4, x^6, \ldots\ldots$



Even func^n

Domain $(x) = (-\infty, \infty) = R$

Range $(y) = [0, \infty) = R^+$

⊛ Poly of Odd Degree

$y = x^3, x^5, x^7, \ldots\ldots$

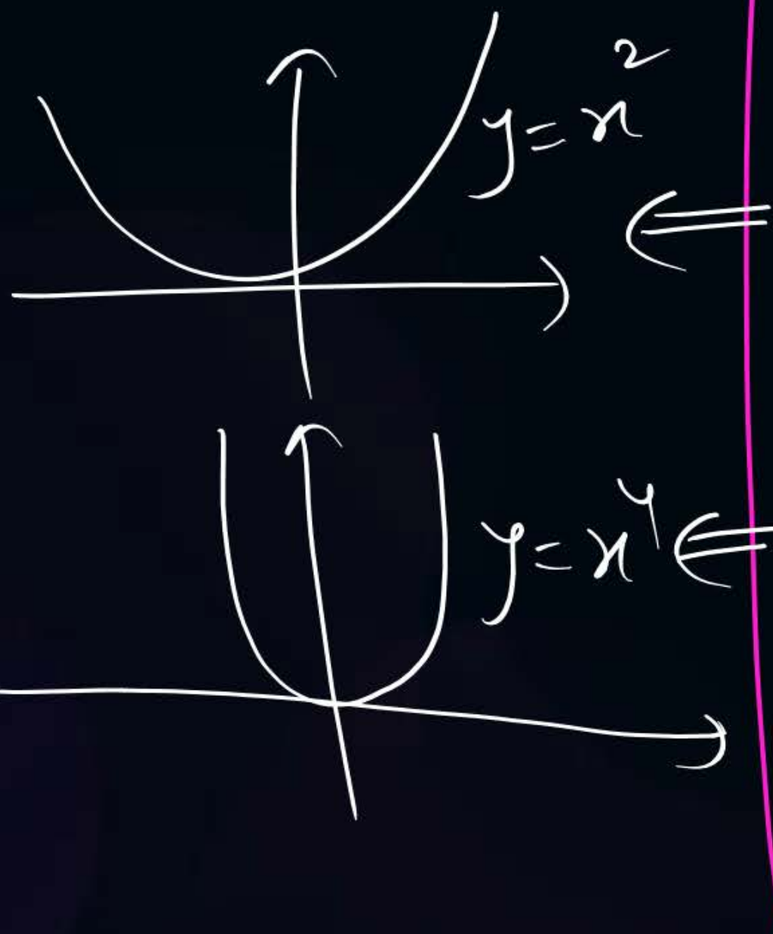


ODD func^n

Domain $(x) = (-\infty, \infty)$

Range $(y) = (-\infty, \infty)$

Doubt



$y=x^2$

...	-3	-2	-1	0	1	2	3	...
	9	4	1	0	1	4	9	...

$y=x^4$

...	-3	-2	-1	0	1	2	3	...
	81	16	1	0	1	16	81	

Rational Func^n

Case I

INFO

$y = \frac{1}{x}, \frac{1}{x^3}, \frac{1}{x^5}, \ldots\ldots$



ODD Func^n

Domain $(x) = R - \{0\}$

Range $(y) = R - \{0\}$

Case II

$y = \frac{1}{x^2}, \frac{1}{x^4}, \frac{1}{x^6}, \ldots\ldots$



EVEN Func^n

Domain $(x) = R - \{0\}$

Range $(y) = (0, \infty)$ i.e. $0 < y < \infty$

ANALYSIS / PODCAST :→ $\frac{\text{Something}}{0} = \text{N.D.}$

M-I

$-\infty \quad -2 \quad -1 \quad -0.1 \quad -0.01 \quad -0.001 \quad 0 \quad 0.001 \quad 0.01 \quad 0.1 \quad 1 \quad 2 \quad \infty$

$\frac{40}{2} = 20$

$\frac{40}{1} = 40$

$\frac{40}{0.1} = 400$

$\frac{40}{0.01} = 4000$

$\frac{40}{0^+} = +\infty$

$\frac{40}{-2} = -20$

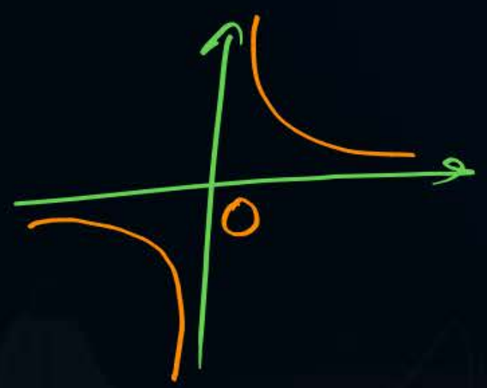
$\frac{40}{-1} = -40$

$\frac{40}{-0.1} = -400$

$\frac{40}{-0.01} = -4000$

$\frac{40}{0^-} = -\infty$

M-II w.k.that $y = \frac{1}{x}$



ie $y(0^+) = +\infty$

& $y(0^-) = -\infty$

ie $\boxed{y(0) = \text{N.D.}}$

Irrational Func^n

INFO

Case I: $y = x^{1/2}$, $x^{1/4}$, $x^{1/6}$ .....

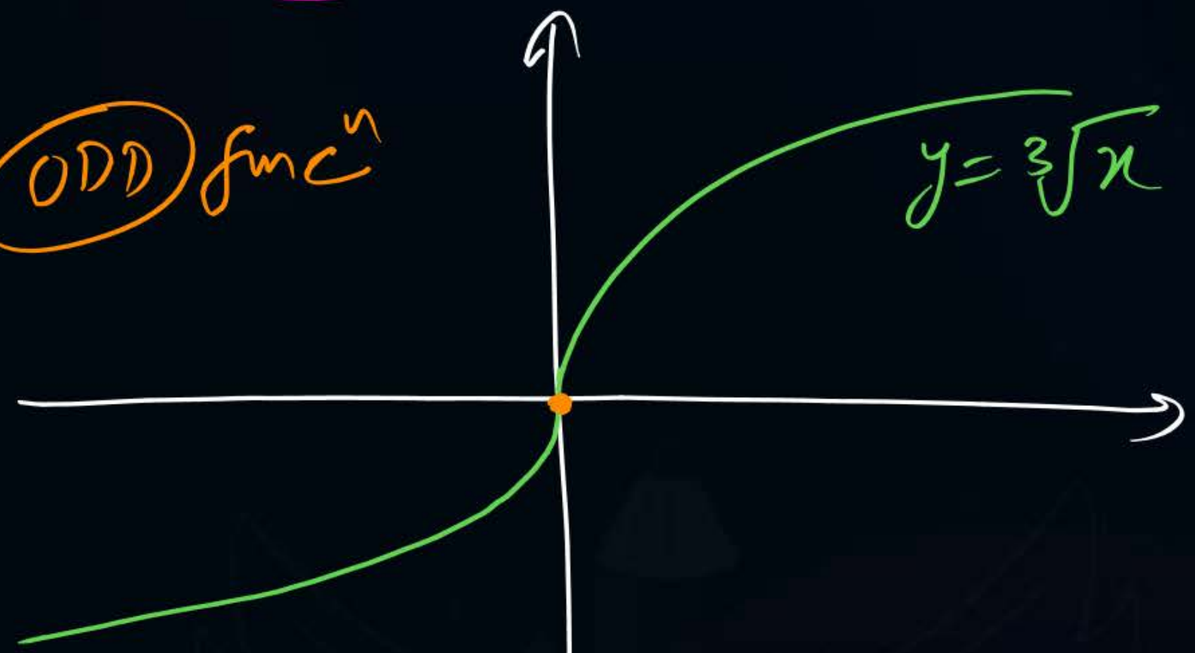


Domain $(x) = [0, \infty)$ i.e. $0 \le x < \infty$

Range $(y) = [0, \infty)$ i.e. $0 \le y < \infty$

Case II: $y = x^{1/3}$, $x^{1/5}$, $x^{1/7}$ .....

ODD Func^n



Domain $(x) = (-\infty, \infty)$

Range $(y) = (-\infty, \infty)$

Even func^n: if $\boxed{f(-x)=f(x)} \Rightarrow f(x)$ is called an Even func^n
& it's Graph is Symmetrical about Y axis

odd func^n if $\boxed{f(-x)=-f(x)} \Rightarrow f(x)$ is called an odd func^n.
& it's Graph is Symmetrical about origin (i.e I $\longleftrightarrow$ III, II $\longleftrightarrow$ IV)

NENO func^n if $f(-x) \neq \begin{cases} f(x) \\ -f(x) \end{cases}$ } then $f(x)$ is called NENO func^n.

it's Graph is neither Symmetrical about Y axis, nor about origin

Note: for the func^n $y = f(x)$

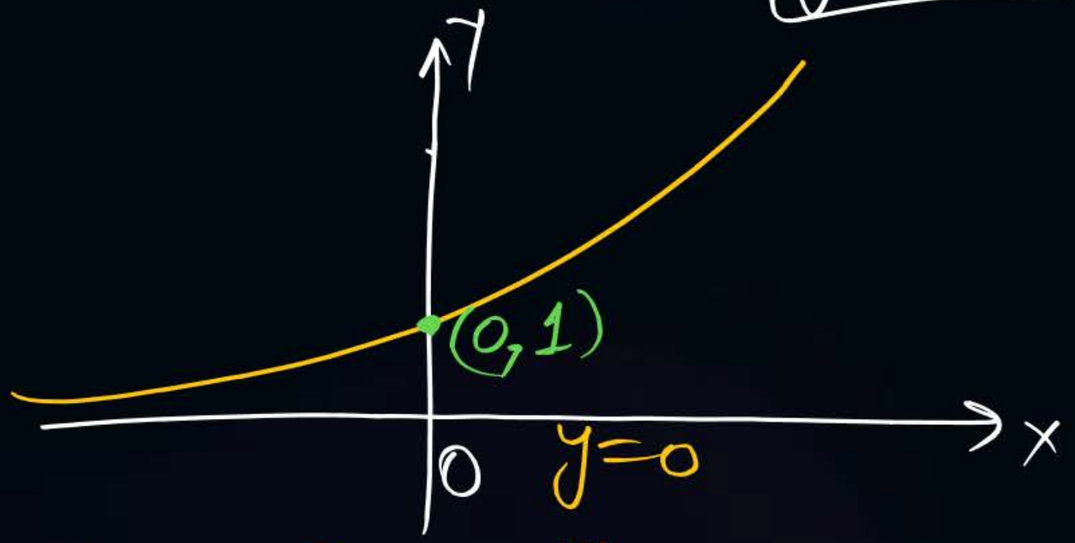
DOMAIN → Set of permissible values of x (for which $f(x)$ is defined) is called Domain. & it lies on X axis

RANGE → Set of permissible values of y is called Range & it lies on Y axis.

⑧ Exponential func^n

$y = e^x$

- Domain(x) = $(-\infty, \infty)$
- Range(y) = $(0, \infty)$ i.e. $0 < y < \infty$
- Nature = NENO



Note: Let $f(x) = e^x$

then $f(-x) = e^{-x} = \frac{1}{e^x} = \frac{1}{f(x)} \neq \pm f(x)$

So $y = e^x$ is a NENO func^n

Note $y = e^x \Rightarrow y = f(x) = e^x$

$y(0) = e^0 = 1$

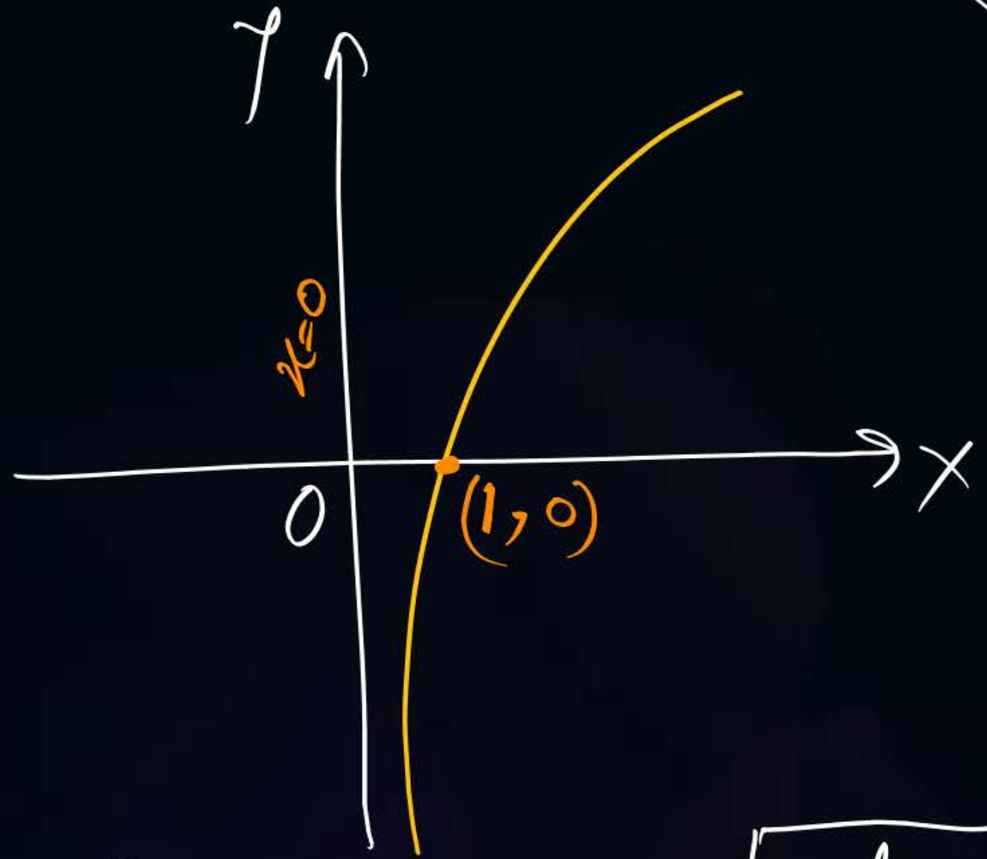
$y(1) = e^1 = e$

$y(\infty) = e^\infty = \infty$

$y(-\infty) = e^{-\infty} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$

(x) log fnc^n → $y = \log_e x$

- Domain $(x) = (0, \infty)$ i.e $x > 0$
- Range $(y) = (-\infty, \infty)$
- Nature = NENO



$y = \log x \Rightarrow y = y(x) = \log_e x$

$y(0) = \log_e 0 \cong -\infty$ = N.D

$y(1) = \log_e 1 = 0$

$y(\infty) = \log_e \infty \cong +\infty$ = N.D.

$\because$ Domain $(x) = (0, \infty) \Rightarrow y = \log x$ is defined only for true values of x.

Doubt →

(x) $\log|x| = \begin{cases} \log(-x), & x < 0 \\ \log(x), & x > 0 \end{cases}$

At $x = -3$, $\log|-3| = \log(3) =$ defined

At $x = 5$, $\log|5| = \log 5 =$ defined


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Pre Requisites for Calculus →

Chapters: 1 to 8, 11th, 14-18,

